# Graphical Research Summary

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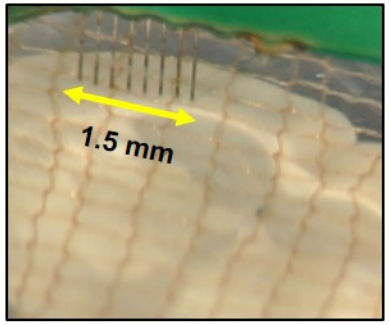
# Contents

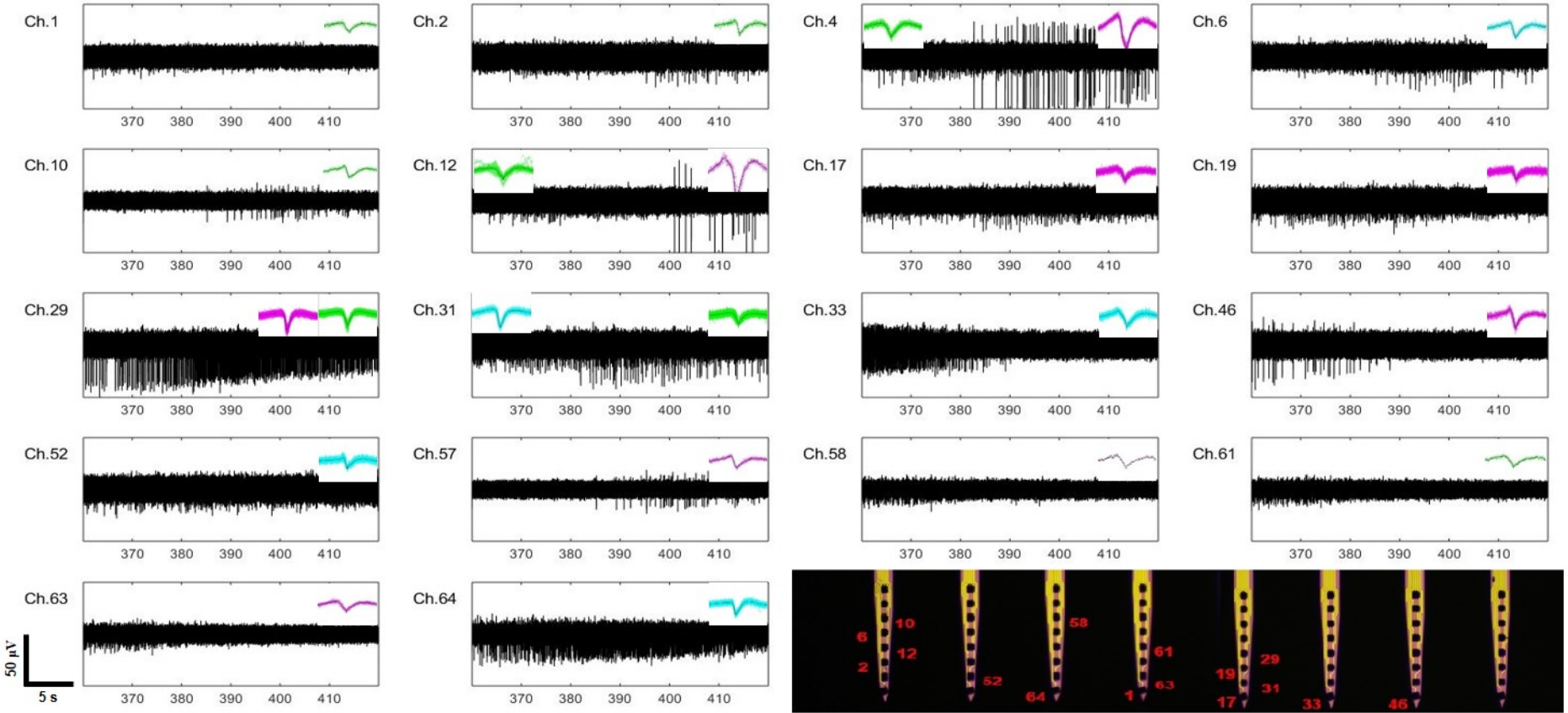
- **Research experience**
  - ***Brain slice electrophysiology system*** (2019-2020, Korea Inst. of Sci. and Tech. – BioMicrosyst. Lab)
  - ***Segregated polymer composites modeling*** (2018-2019, Korea Univ. – Communication Systems Lab)
  - ***Retinal connectomics project – EyeWire*** (2015, Princeton Univ. – Comp. Neuroscience Lab)

- **My Doctoral Goal**
  - ***To implement closed-loop control using noninvasive brain stimulation methods***

# Brain slice electrophysiology system
## (Korea Institute of Science and Technology, 2019 – 2020)

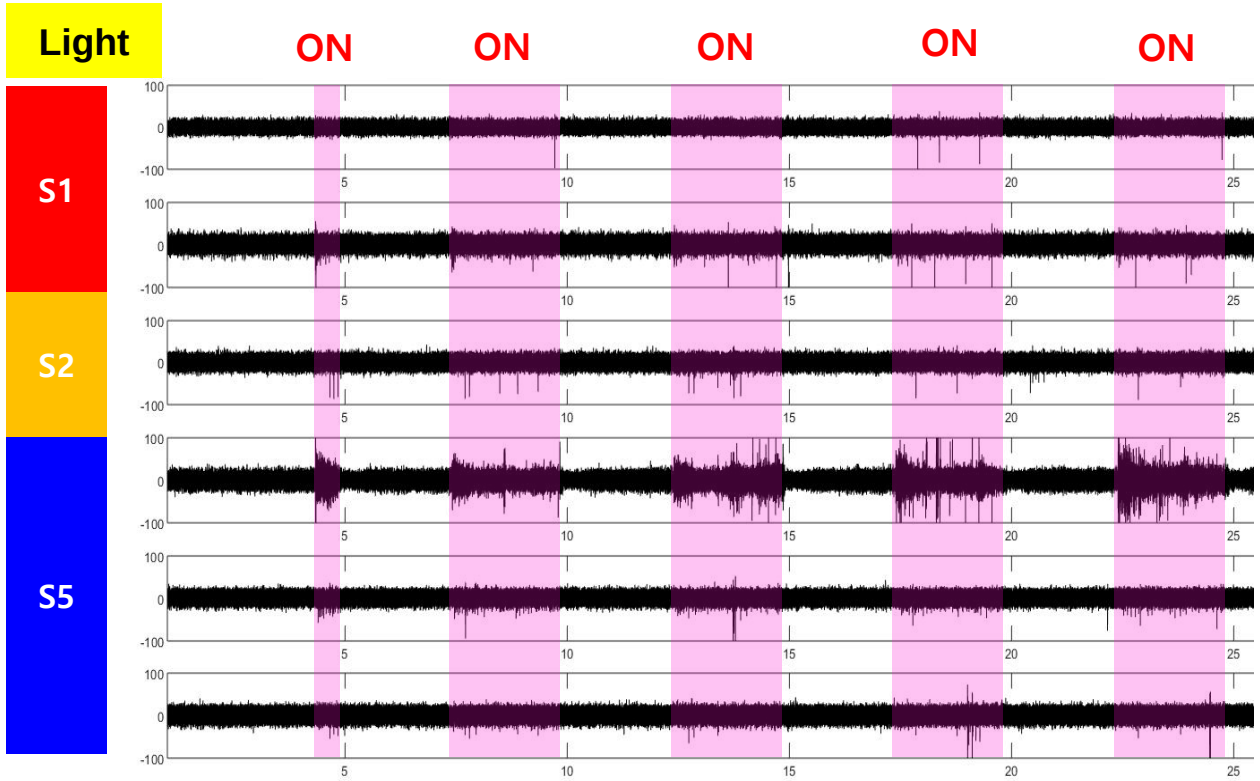
- Extracellular recording probe
  - 64 electrodes
  - 8 shank – 1.5/3 mm span
- Action potentials – cerebral cortex, 400 μm thick
  - Observed from as many as **18 electrodes** (w/ KCl injection)
  - Recorded for as long as **4 h 20 min**

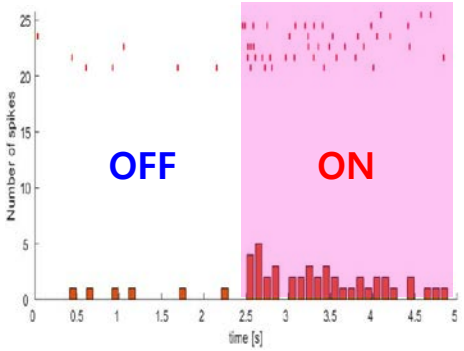


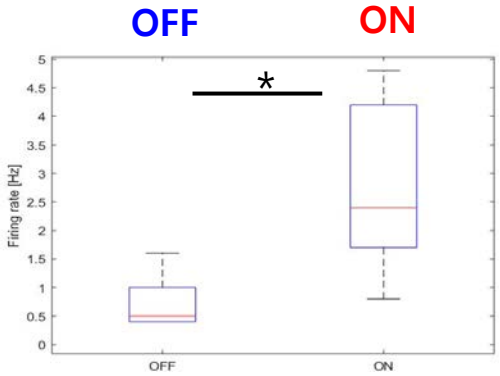
# Brain slice electrophysiology system
## (Korea Institute of Science and Technology, 2019 – 2020)

- **Targeted recording** from ventral HPC (CA1 w/ Schaffer collateral stim.)
  - 330 μm thick hippocampal slice (Thy1-ChR2-YFP)

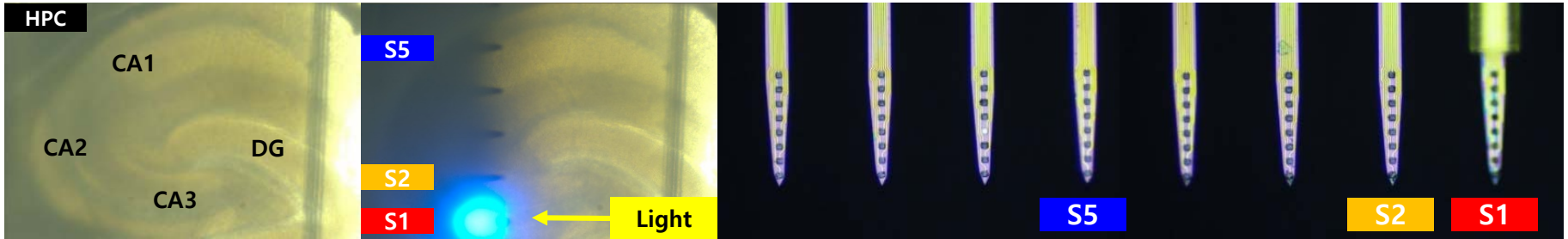


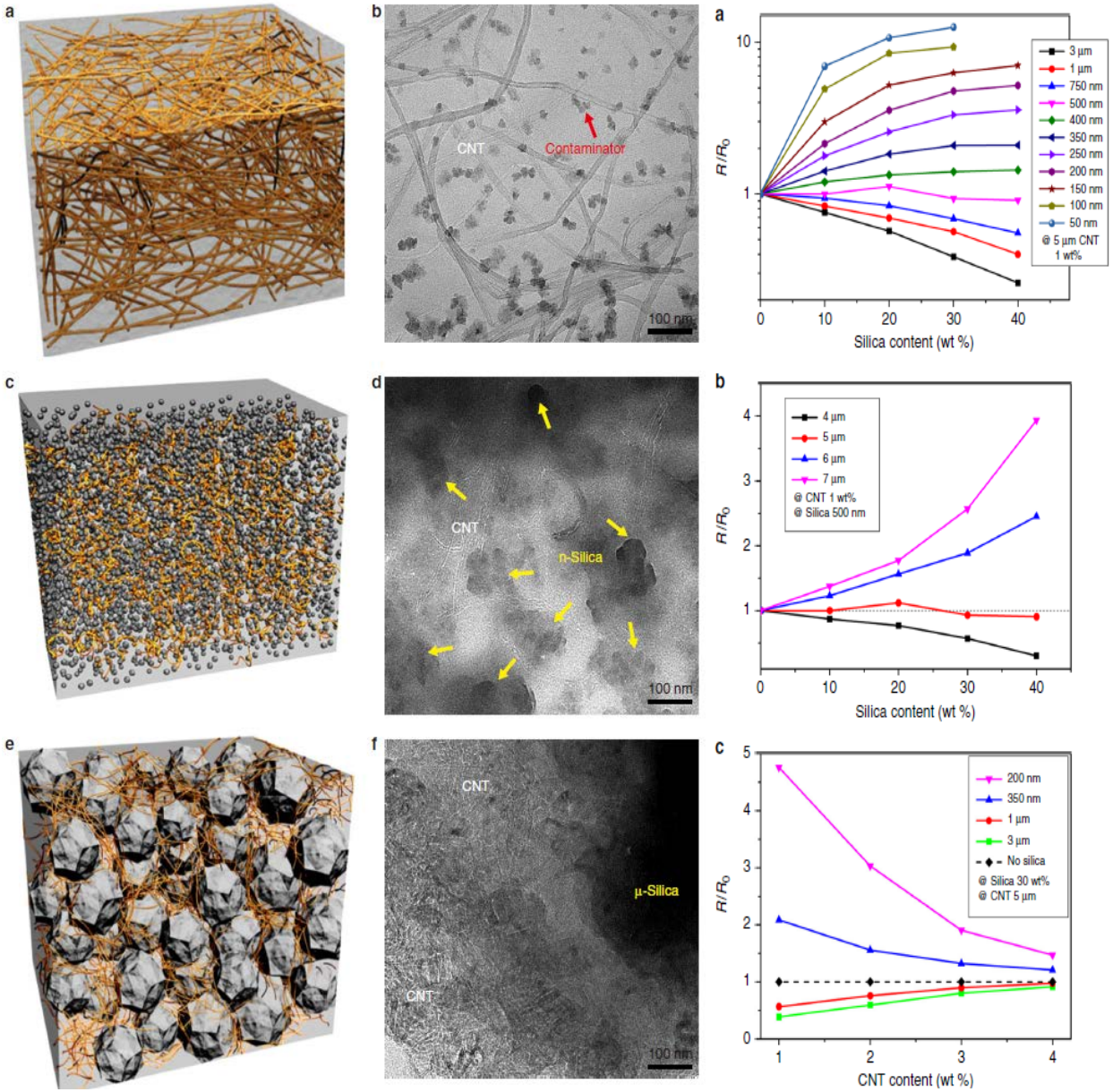

Raster plot




Firing rate

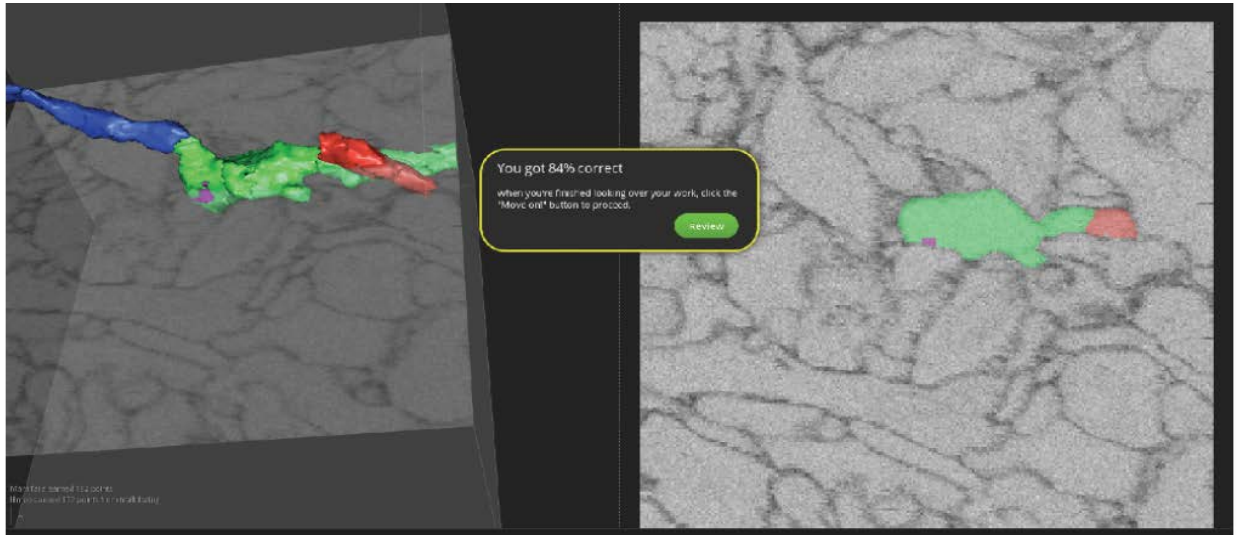


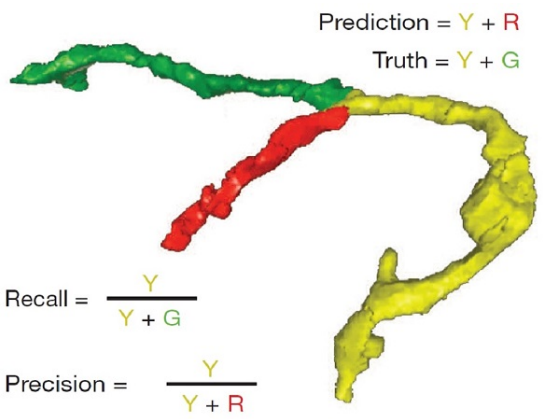
# Segregated Polymer Composites Modeling
## (Korea University, 2018 – 2019)

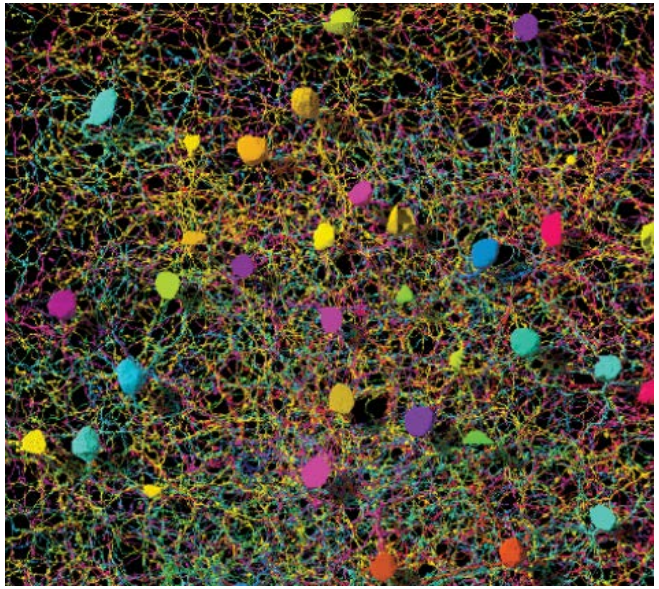



- **Aim:** to model the electrical conductance properties of polymer composites with segregated structures
- Raising microscale fillers percentage – decreases conductivity due to the prevention of the interconnections between CNTs.
- Raising nanoscale fillers percentage – enhances conductivity based on exclude volume theory
- **Our approach:** simulated the polymer composite in MATLAB and measured the electrical resistivity through Monte Carlo simulation to explain the results using Voronoi geometry and percolation theory
- **Publication:** *Nature Communications (June, 2019)*
- **My contribution:** built the 3D CNT network simulator and revised the manuscript together

# Retinal Connectomics Project – EyeWire
## (Princeton University, Jan – Feb 2015)



- **Aim:** to reconstruct 3D models of neurons from electron microscopy (EM) images via machine learning and crowd sourcing
- **Our approach:** Transform the labeling process of 2D EM images into an entertainment software to data generated by vast online users. The collected user data was used to train the deep convolutional network which classifies retinal cells.
- **Result:** mapped structures of over 3000 retinal neurons with the help of 2.7 million participants
- **Details:** https://blog.eyewire.org/about/
- **My contribution:** improved the data assessment algorithm of the starburst amacrine cell by computing its stratification profile via using a support vector machine